

RSME - Vendor Document / Drawing Review Stamp (IDC)					
EPC for Upgradation of Facilities at SE Field					
Project	Date	Discipline	Req.	Ref. No.	Code / Action
1		Pre			
2		EM	✓		
3		Process			
4		Piping	✓		
5		Pipeline			
6		Mechanical			
7		Civil / Structure	✓		
8		Electrical	✓		
9		Inst / Com.	✓		
10		Telecom			
11		Materials			
12		HVAC			
13		QA/QC	✓		
14		HSE	✓		
15		PLANNING			
16		Construction			
17		Procurement			
18		Coating / Painting / Insulation			
Notes: Code 1: No comments, Issue to COMPANY for review / Approval Code 2: Approved - Subject to incorporation of comments / Approved with Comments-Resubmit Code 3: Review and Resubmit - Work can NOT Proceed Code 4: Rejected Code 5: Reviewed for Information Code 6: TOL/Revised/Resubmit					
Legends				Required (Req.)	✓ No



Abu Dhabi Company for Onshore Petroleum Operations Ltd.
(ADNOC ONSHORE)

EPC FOR UPGRADATION OF FACILITIES AT SE FIELD

AGREEMENT NO.: 4700022908
COMPANY PROJECT NO.: P30377

DATASHEET & CATALOGUES FOR HSSD AT 33KV
SUBSTATION in SHAH CDS

P30255-29-01-40-4602

This document is prepared for the following projects and the project number P30377 is for common deliverables.

PROJECT DESCRIPTION	PROJECT NUMBER
EPC for Upgradation of Facilities at SE Field	P30377
Replacement of Aged Electrical Equipment at SE Field	P30255
Replacement of Existing ESP Power Skids (24 Nos) at SHAH	P29179
Requirement of WHCP to operate downhole SSSV at SHAH	P29175
Replacement of SDH Network and Telecom UPS at ASAB / SAHIL	P14429
Physical Security recommendations Project at BHU/ SE	P30297
Replacement of Water Disposal GRE Line at SHAH	P29171

A	27-10-2025	MUTHUGANAPATHY	THIRUVENGATESH	JAYARAJ	Issued for Review
REVISION	DATE	PREPARED	REVIEWED	APPROVED	DESCRIPTION

THIS DOCUMENT IS INTENDED FOR USE BY COMPANY AND ITS NOMINATED CONSULTANTS, CONTRACTORS, MANUFACTURERS AND SUPPLIERS

EPC CONTRACTOR: ROBTSTONE (ME) LLC	VENDOR/ SUB-CONTRACTOR:

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ADNOC ONSHORE AGREEMENT NO. 4700022908	ADNOC ONSHORE PROJECT NO. P30377	REV. A
		DATE: 27-10-2025

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1 GENERAL

1.1 Introduction

ASAB, SAHIL & SHAH fields are the oil fields located at south east of Abu Dhabi city and are operated by Abu Dhabi COMPANY for Onshore Petroleum Ltd (COMPANY). SAHIL, ASAB, & SHAH (SAS) fields are located in the desert area approximately 100 km, 160 km and 220 km, respectively from the Abu Dhabi city - see Figure 1.

Bu-Hasa field is located 200 kilometers south west of Abu Dhabi as shown in figure 1. It is ADNOC Onshore's most western producing field. It was discovered in 1962 and was put on production in May 1965. In Bu-Hasa currently there are three Remote Degassing Stations (RDS-1/2/3) and two satellite stations BQ & HW.

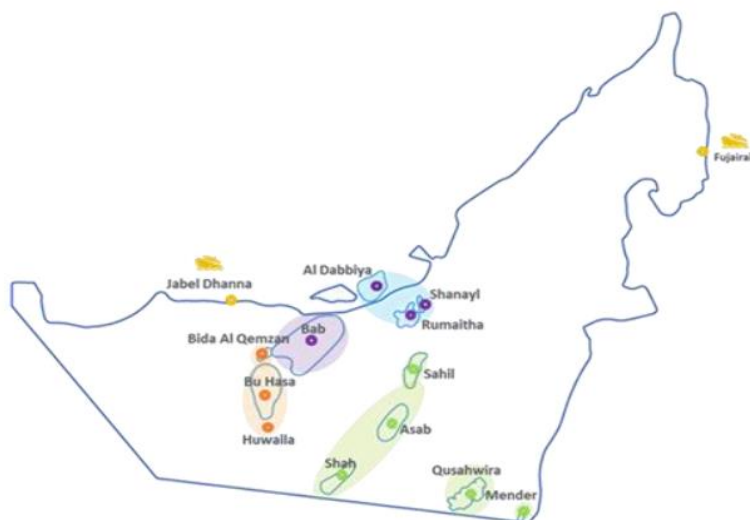


Figure-1: ADNOC ONSHORE FIELDS

The COMPANY awarded the Engineering, Procurement and Construction works for "Upgradation of Facilities at SE Fields" Project to M/s ROBT STONE (ME) LLC- as EPC-contractor.

1.2 Scope

The Scope of Work for EPC CONTRACTOR to carry out Modification Works at South East & Bu Hasa to implement for the purposes of maintaining continuous improvement for the existing facilities and concerns raised by operations, it is intended carrying modifications to improve the operability and safety of the existing facilities; consisting of modifications and upgradations

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of existing facilities, enhance the fault recovery of transmission system and enhance the security system.

The Scope of work comprises the following packages:

- i. Package 1: Replacement of Aged Electrical Equipment at SE Fields.
 - a. ASAB
 - b. SAHIL
 - c. SHAH
 - d. Sub package 1: Replacement of existing ESP power skids(24Nos) at Shah
 - e. Sub package 2: Requirement of WHCP to operate downhole SSSV in Shah
- ii. Package 2: Replacement of SDH Network and Telecom UPS at Asab and Sahil.
- iii. Package 3: Physical Security Recommendation at Bu Hasa and SE Fields.
- iv. Package 4: Replacement of Water Disposal GRE Lines at SHAH.

1.3 Purpose of this Document

The Purpose of this document is to define the Datasheet & Catalogue for HSSD System supplying by Tyco Fire & Security UAE LLC for the project “EPC FOR UPGRADATION OF FACILITIES AT SE FIELD”.

1.4 Definition and Glossary

CONTRACT	:	4700022908
PROJECT NUMBER	:	P30377
COMPANY	:	Abu Dhabi Company for Onshore Petroleum Operations Ltd (ADNOC Onshore)
EPC CONTRACTOR / CONTRACTOR	:	Robt. Stone ME (RSME)
SUBCONTRACTOR	:	Agency / Entity that will be engaged by Contractor in agreement with Company to perform specific work during EPC stage
MANUFACTURER / SUPPLIER / VENDOR	:	The person, firm, corporation including their assignees To whom the purchase order is placed by CONTRACTOR.
PROJECT	:	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD

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SHALL	:	The use of the word “shall” indicate a mandatory requirement
SHOULD	:	Indicates a strong recommendation to comply with the requirement the document
MAY	:	Indicates preference course of action

1.5 Abbreviation

Abbreviations	:	Description
ADNOC	:	Abu Dhabi National Oil Company (ADNOC Onshore)
PROJECT	:	EPC FOR UPGRADATION OF FACILITIES AT SE FIELD
HSSD	:	High Sensitivity Smoke Detection
IFC	:	Issued for Construction
IFR	:	Issued for Review
CPU	:	Central Processing Unit
AC	:	Alternating Current
DC	:	Direct Current
LCD	:	Liquid Crystal Display
LED	:	Light Emitting Diode
IP	:	Ingress Protection
UL	:	Underwriters Laboratories
UPS	:	Uninterruptable Power Supply
V	:	Voltage
W	:	Watt
mA	:	Milli-Amps
Mm	:	Milli-Meter
Hz	:	Hertz
°C	:	Degree Celsius

1.6 Reference Code and Standards

The services shall be carried out in accordance with good engineering practices, latest revisions of all COMPANY specifications and recognized international codes of practices, standards and specifications along with their addenda applicable for the PROJECT.

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1.6.1 Project Specification Documents

Include SHAH CDS SS F&G Layout, HSSD layout, HSSD flow calculation, Electrical Equipment layout and other HSSD related deliverable in section 1.6.1

Document Number	Title
P30255-29-99-93-1603	MR FOR HIGH SENSITIVITY SMOKE DETECTION SYSTEM_Rev-A
P30255-29-99-39-1601	SPECIFICATION FOR HIGH SENSITIVITY SMOKE DETECTION SYSTEM_Rev B

1.6.2 Project Vendor Documents

Document Number	Title
P30255-29-01-47-4603	HSSD GA & WIRING DRAWING FOR 33KV SUSBATATION in SHAH CDS

1.6.3 AGES and ADNOC Onshore Standards

Document Number	Title
AGES-SP-04-003 Rev 1	Fire and Gas System Specification
AGES-PH-03-002 Rev 1	Fire & Gas Detection and Fire Protection System Philosophy

1.6.4 Shell DEP's & Specification

Shell DEPs Version 43 to be followed.

Document Number	Title
DEP 32.30.20.11	Fire, Gas and Smoke Detection Systems

1.6.5 International Codes and Standards

The International Codes, Standards and Recommended Practices which shall be utilised by CONTRACTOR for the execution of the works are referenced in the project documents.

Document Number	Title
ISO 9001	Quality Managements System Requirements
BS EN 54	Fire Detection and Alarm Systems
BS EN 54-20	Fire detection and fire alarm systems – Part 20: Aspirating smoke detectors

1.6.6 Order of Precedence

All design and construction shall be performed in accordance with the Specifications, Standards, Codes, Regulations, latest Shell DEPs, etc. listed in the Contract.

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In any areas of conflict, the order of precedence in the use of the revisions of the above as at the Effective Date of the Contract shall be as follows:

- a. The Laws, Standards and Regulations of the United Arab Emirates.
- b. ADNOC Code of Practice, HSE manuals and policies.
- c. Project Scope of Work.
- d. Project specific data sheets / specifications including ADNOC AGES Standards.
- e. COMPANY Procedures and Codes/Standards
- f. COMPANY Specifications and Engineering Practices.
- g. COMPANY amendments and supplements to Shell DEPs.
- h. Shell DEPs (v43).
- i. International oil and gas industry Codes, Standards and Recommended Practices (all where specified in the above or, where none of the above is applicable, as proposed by Engineering Contractor and approved by COMPANY).
- j. Internationally recognized sound practices in oil and gas industry. In cases of conflict between documents in the same level of the hierarchy, the most stringent requirement shall apply. In such cases, CONTRACTOR shall provide its interpretation in writing of the most stringent requirement for COMPANY's approval utilizing a Technical Query sheet. In all cases of conflict, COMPANY's decision shall be final.

International Oil & Gas Industry Codes, Standards, and Recommended practices (all where specified in above or, where none of the above is applicable, as proposed by the CONTRACTOR and approved by ADNOC Onshore)

In cases of conflict between documents, the most stringent requirement shall apply. In such cases CONTRACTOR shall provide its interpretation in writing of the most stringent requirement for ADNOC Onshore's Approval utilizing a Technical Query sheet (to be obtained from ADNOC Onshore). In all such cases of conflict ADNOC Onshore's decision shall be final without any time and cost impact.

2 QUALITY ASSURANCE

EPC Contractor shall operate a Quality Management System in compliance with ISO 9001 Ed 2015, and AGES-GL-13-001 Rev 01 in all stages of Project.

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Quality Control / Inspection & test plan shall be prepared as per Criticality rating & inspection class of the materials in compliance to Spec AGES-SP-13-001 Rev 01 & AGES-SP-13-002 Rev 01, for Procurement Inspection and Certification Requirement in Projects.

COMPANY & EPC CONTRACTOR, or Appointed Representative, reserves the right to carry out Quality Audits as defined in EPC Contractor Project Quality Plan to ensure that the Quality System requirements are being met.

3 SAFETY

CONTRACTOR shall prepare and maintain a project specific HSE plan that identifies and schedules all HSE activities of the project clearly identifying the HSE deliverables for all project phases in accordance with COMPANY Guidelines for the Preparation of HSE Plans This shall be a live document and shall be updated to include additional studies as well as schedule for all HSE activities during each stage by the CONTRACTOR.

4 ENVIRONMENT (UAE WEATHER)

4.1 Climatic Conditions

Wind		
i)	Prevailing Direction	From N-NW
ii)	Mean Speed	8 m/s
iii)	Basic Wind Speed (3 second gust)	44.7 m/s (as per ASCE-7 definition)
iv)	Basic Wind Speed (Hourly Mean Wind Speed)	26 m/s (as per BS 6399 definition)
v)	Airborne Particles	Saliferous, sulfurous, dust and sand with frequent sandstorms
Temperature		
i)	Maximum Solar	85 °C
ii)	Maximum Shade	58 °C
iii)	Average Shade Summer	36 °C
iv)	Average Shade Winter	22 °C
v)	Average Shade Yearly	28 °C
vi)	Minimum Shade	4 °C
vii)	Maximum Soil at 1m depth	38 °C
viii)	Minimum Soil at 1m depth	13 °C

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Humidity		
i)	Relative Maximum at 43 °C	95 %
ii)	Relative Maximum at 54 °C	60 %
Solar Radiation		946 W/m2 1150-1250 W/m2 for O/HL design
Rainfall		
i)	Frequency	Infrequent
ii)	Maximum	51 mm/year
iii)	Minimum	Trace
iv)	Highest rate	January-April, 25 mm in 24 hrs.
Dew		Heavy (daily)
Mists		Early morning mists causing Evaporative cooling loss.
Soil Thermal Resistivity		2.5 °C m/W

4.2 Equipment Design Temperature (Under Shed or Indoor)

Electrical Equipment Design Temperature	55 °C (40 °C indoor)
Motors, Generators	55 °C
Instrument Equipment	60 °C
Electrical buried Cables (at 1m depth)	40 °C
Electrical Cables in Air, Duct or Tray	55 °C
Air Cooler – non-process	55 °C

4.3 A/C Design Conditions

Outdoor design temperature (Summer)	50 °C dry bulb/ 30 °C wet bulb
Outdoor daily range	15 °C
Outdoor design temperature (Winter)	6 °C dry bulb
Indoor Design Conditions	24 °C dry bulb/ 50% R. Humidity
Ambient Temperature for A/C selection	52 °C

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5 SPECIFICATION [ENGINEERING] REQUIREMENTS

EPC CONTRACTOR will develop and ensure that all engineering and design work required for the Project scope is performed, implemented, and properly recorded on documents in a professional manner, and in accordance with the Contract requirements and oil and gas industry practices

Do not leave blank spaces in the document-typical comment.

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6 DATASHEET AND CATALOGUES FOR HSSD

HIGH SENSITIVITY SMOKE DETECTION			
GENERAL	1.	Area Classification	Safe Area
	2.	Quantity & Device Tag	02 Nos
	3.	Device Tag	29-01-HSSD-5802-01 & 29-01-HSSD-5802-02
TECHNICAL DATA	4.	Detection Method	Flair Detection Technology
	5.	Type	Aspirating Smoke Detector - Xtralis VES-A10-P.
	6.	Incoming Power Supply	240 VAC 50 Hz UPS supply (Incomer to PSU)
			18-30 VDC from Power Supply Unit to HSSD
	7.	Display	3.5" LCD Touchscreen Display
			Seven (7) Status LEDs: (for Fire 1, Fire 2, Action, Alert, Disabled, Fault, Power)
			Push Button - RESET/ DISABLE/ ACKNOWLEDGE
			Ten (10) Icons: [for various Fault categories (8), Smoke Level and Adaptive Scan Threshold Indication (1), Detector Normal Indication (1)]
			Four (4) Screens: Main screen, Sector Status screen, Airflow screen, Status screen
	8.	Dimension (WxHxD)	350 x 225 x 135 (mm)
	9.	Weight	4.4 kg
	10.	Ambient Temperature	0°C to 39°C
	11.	Operating Humidity	5% to 95% RH, non-condensing
	12.	Mounting	Surface mountable
	13.	Area Coverage	2000m2
	14.	IP rating	HSSD Unit IP40
			Enclosure IP54
	15.	Relays	12 programmable relays
	16.	Cable Termination	Screw terminal blocks (0.2–2.5 Sq.mm)
	17.	No. of Pipe Inlets	4 Nos
	18.	Pipe Length	Linear 280m max.
			Branched 560m max.
	19.	Max allowed Transport Time	120 seconds
	20.	Power Consumption	Quiescent 16.3 W (Max) @ 24 VDC
			Alarm 17.3 W (Max) @ 24 VDC
	21.	Event Log	20,000
	22.	Sensitivity Range	0.005 to 20% obs/m
	23.	Self-Testing and Diagnostics	Confirmed Self-Testing and Diagnostic feature

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HIGH SENSITIVITY SMOKE DETECTION			
	24.	Response time	=<2 minutes
	25.	Aspirator Fan	Integrated aspirator available inside HSSD
ACCESSORIES	26.	Tag Plate	Traffolyte Tag Plate (As per ADNOC Standard, ES-30-99-00-0001-1)
HSSD ENCLOSURE	27.	Enclosure Protection	IP54
	28.	Material	FRP/GRP
	29.	Finish	RAL 7035 (Manufacturer Standard)
	30.	Mounting	Surface Mountable
	31.	Cable Entry	Bottom Entry
POWER SUPPLY UNIT (PSU)	32.	Make / Model	PHOENIX / 2907161
	33.	Type	TRIO-UPS/1AC/24DC/10
	34.	Output Voltage	22.5 – 29.5 VDC (24 VDC Nominal)
	35.	Output Current	10A
	36.	Power Dissipation	16W @ 230 VAC
	37.	Temperature	-25°C to +70°C
	38.	Humidity	95% RH, no-condensation
BATTERY	39.	Battery Mounting	Battery is placed in HSSD Enclosure
	40.	Battery Size	12V 21Ah (2 Nos)
	41.	Back-Up Hours	Standby - 24 hours Alarm - 30 minutes
	42.	Battery Make / Model	POWER SONIC / PS-12210
APPROVALS	43.	Approvals (HSSD)	UL, VdS, EN54-20
PURCHASE	44.	Manufacturer	XTRALIS
	45.	Model No	VES-A10-P
HSSD PIPES & FITTINGS			
GENERAL	46.	Area Classification	Safe Area
	47.	Quantity	1 Lot per HSSD panel (as per Design drawings)
TECHNICAL DATA	48.	Material	ABS
	49.	Internal Diameter	22 mm
	50.	External Diameter	25 mm
	51.	Color	Red
SPECIAL TOOLS	52.	Wire Burn Test Box	VTT-10000
	53.	Smoke Test Wire	251-001
PURCHASE	54.	Manufacturer	XTRALIS

Ensure the battery capacity based on the HSSD panel & accessories load as required, provide calculation.

Include approvals such as UL, FM, LPCB, VDS etc. for the HSSD pipes & fittings.

Notes: 1. Material Criticality Rating & Inspection Class as per AGES-SP-13-001 & 002 shall be 1
2. Material Test Certificate and product certificate to BS EN 10204 shall be 3.1

VESDA-E VES

VES-A00-P, VES-A10-P



The VESDA-E VES is similar to the flagship VESDA-E VEP aspirating smoke detector but also includes a valve mechanism in the inlet manifold and software to control the airflow from the four Sectors (pipes). This configuration enables a single zone to be divided into four separate sectors, for example, distinguishing between separate aisles within a data room. The VES enables the user to locate the source of smoke by identifying the first sector to reach the Alert level. The detector then continues to sample from all sectors to monitor fire growth and will report separate alarm levels for each sector. The VES provides four individually configurable alarm levels (Alert, Action, Fire 1 and Fire 2) for each sector allowing optimum protection in a wide range of applications. Built on the Flair detection technology and years of application experience, VES detectors achieve consistent performance over their lifetime via absolute calibration. In addition, the VES delivers a range of revolutionary features that provide user value.



How it Works

The VES draws air from all sectors in use. If the smoke level reaches the Adaptive Scan Threshold, the VES quickly scans each sector to identify which sector is carrying smoke. The first sector to reach the Alert level is designated as the First Alarm Sector (FAS) and this sector is signalled to the user. If two or more sectors reach the Alert level then, the sector with the highest smoke concentration is designated as the First Alarm Sector (FAS). Once Fast Scan is completed and the FAS identified, the VES continues to closely monitor all four sectors to track fire growth and maintain full protection of the area.

Flair Detection Technology

Flair is the revolutionary detection chamber that forms the core of the VESDA-E VES, providing higher stability and increased longevity. Direct imaging of the sampled particles using a CMOS imager combined with multiple photo-diodes allows better detection and fewer nuisance alarms.

The VES Display

The VES display Home page has a bar graph to indicate the smoke level and adaptive scan threshold. Fault icons are also included to indicate various fault conditions. When the adaptive scan threshold is exceeded the VES display automatically transitions to the Sector status page to indicate the smoke level and alarm level per sector. If alarms are configured as latched then alarm indication per sector will be retained until Reset is applied. The VES display can only return to the Home page under user control.

Installation, Commissioning and Operation

VESDA-E VES is equipped with a powerful aspirator that enables the use of 560 m (1,837 ft) of total pipe length. Out of box operation is made possible with AutoConfig which allows airflow normalisation and AutoLearn Smoke and Flow to be initiated from within the detector. VES is fully supported by Xtralis VSC and ASPIRE software applications which facilitate ease of pipe network design, system commissioning and maintenance.

VESDAnet™

VESDA devices communicate on VESDAnet which provides a robust bidirectional communication network allowing continued redundant operation even during single point wiring failures. VESDAnet enables primary reporting, centralized configuration, control, maintenance and monitoring.

Ethernet connectivity

VESDA-E detectors offer connectivity to corporate networks via Ethernet, allowing for devices installed with Xtralis monitoring and configuration software to connect to the detector.

Backward Compatibility

VESDA-E VES is compatible with existing VESDA installations. The detector occupies the same mounting footprint, pipe, conduit and electrical connector positioning as VESDA VLS. VES is also compatible with existing VESDAnet installations allowing monitoring of both VESDA-E and legacy detectors via the latest VSC and VSM4 applications.

Features

- Sector addressability for up to four sectors
- Adaptive scan threshold
- Flair detection technology delivers reliable very early warning in a wide range of environments with minimal nuisance alarms
- Short wavelength laser-based detection:
 - High sensitivity from small particle light scattering
 - No drift compensation required since focused light directed at target gives low backgrounds
 - High stability with temperature and time
- Multi stage filtration and optical protection with clean air barriers ensures lifetime detection performance
- Four configurable alarm levels per sector and a wide sensitivity range deliver optimum protection for the widest range of applications
- Intuitive LCD display provides instant status information for immediate response
- Flow fault thresholds per port accommodate varying airflow conditions
- Smart on-board filter retains dust count and remaining filter life for predictable maintenance
- Extensive event log (20,000 events) for event analysis and system diagnostics
- AutoLearn™ smoke and flow for reliable and rapid commissioning
- Backward compatible with VLS and VESDAnet
- Ethernet for connectivity with Xtralis software for configuration, secondary monitoring and maintenance
- USB for PC configuration, and firmware upgrade using a memory stick
- Two programmable GPIOs (1 monitored) for flexible remote control
- Field replaceable sub-assemblies enable faster service and maximum uptime

Listings / Approvals

- CSFM
 - FM
 - ActivFire
 - VdS
 - NF-SSI (www.marque-nf.com)
 - CE
 - UKCA
 - EN 54-20, ISO 7240-20
 - Class A (40 holes / Fire 1 = 0.067% obs/m)
 - Class B (80 holes / Fire 1 = 0.085% obs/m)
 - Class C (100 holes / Fire 1 = 0.251% obs/m)
- Classification of any configuration is determined using ASPIRE.

Regional approvals listings and regulatory compliance vary between product models. Refer to www.xtralis.com for the latest product approvals matrix.

Highlight/Mark the selected model, applicable parameters/criteria etc. in the specification.

VESDA-E VES

TECHNICAL SPECIFICATIONS



Specifications

Supply Voltage	18-30 VDC (24 V Nominal)					
Power Consumption @ 24VDC	VES-A00-P			VES-A10-P		
Aspirator Setting	1	5	10	1	5	10
Power (Quiescent)	7.5 W	9.8 W	15.4 W	8.0 W	10.4 W	16.3 W
Power (In Alarm)	8.4 W	10.8 W	15.8 W	9.2 W	11.3 W	17.3 W
Dimensions (WHD)	350 mm x 225 mm x 135 mm (13.8 in x 8.9 in x 5.3 in)					
Weight	4.7 kg (10.4 lbs)			4.8 kg (10.6 lb)		
Operating Conditions	Ambient: 0°C to 38°C (32°F to 100°F) Sampled Air: -20°C to 60°C (-4°F to 140°F) * Humidity: 5% to 95% RH, non-condensing					
Area Coverage	2,000 m ² (21,520 sq. ft)					
Min. Airflow per Pipe	20 l/m					
Pipe Length (Linear)	280 m (919 ft)**					
Pipe Length (Branched)	560 m (1,837 ft)**					
Pipe Lengths Depending on No. of Pipes in Use	2 Pipes		3 Pipes		4 pipes	
	100 m (328 ft)		80 m (262 ft)		70 m (230 ft)	
No. of Holes (A/B/C)	40/80/100**					
Computer Design Tool	ASPIRE					
Pipe	Inlet: External diameter 25 mm or 1.05 in (3/4 in IPS) Exhaust: External diameter 25 mm or 1.05 in (3/4 in IPS) via adaptor					
Relays	12 programmable relays (latching or non-latching states) Contacts rated 2 A @ 30 VDC (Resistive)					
IP Rating	IP40					
Cable Access	4 x 26 mm (1.02 in) cable entries					
Cable Termination	Screw Terminal blocks 0.2–2.5 sq mm (24–14 AWG)					
Measurement Range	0.000 to 32% obs/m (0.0000 to 11.09% obs/ft)					
Sensitivity Range	0.005 to 20% obs/m (0.0015% to 6.575% obs/ft)					
Threshold Setting Range	Alert: 0.005% to 2.0% obs/m (0.0015% to 0.614% obs/ft) Action: 0.005% to 2.0% obs/m (0.0015% to 0.614% obs/ft) Fire1: 0.010% to 2.0% obs/m (0.0030% to 0.614% obs/ft) Fire2: 0.020% to 20.0% obs/m (0.0061% to 6.575% obs/ft)					
Software Features	Event log: Up to 20,000 events Smoke level and alarm threshold levels, user actions, alarms and faults with time and date stamp AutoLearn: Detector learns Alarm Thresholds and Flow Fault thresholds by monitoring the environment.					

* Sampled Air temperature shall reach Detector Ambient temperature upon entry into Detector. Refer to Xtralis Design Guides & Application Notes for sampled air pre-conditioning.
** Subject to agency confirmation.

Ordering Information

Ordering Code	Description
VES-A00-P	VESDA-E VES with LEDs, Plastic Enclosure
VES-A10-P	VESDA-E VES with 3.5" Display, Plastic Enclosure
VES-A00-P-NF	VESDA-E VES with LEDs, Plastic Enclosure - NF
VES-A10-P-NF	VESDA-E VES with 3.5" Display, Plastic Enclosure - NF
VKT-855	VESDA-E VES Demo Kit

Spare Parts

VSP-955	VESDA-E VES Scanner Manifold	VSP-964-03	VESDA-E Smoke Detection Chamber - MK3
VSP-955-04 *	VESDA-E VES Scanner Manifold	VSP-964-04 *	VESDA-E Smoke Detection Chamber - MK4
VSP-960	VESDA-E Mounting Bracket	VSP-965	VESDA-E Sampling Module
VSP-961	VESDA-E Exhaust Adaptor US	VSP-968	VESDA-E VES-A00-P Front Cover - Plastic - LEDs
VSP-962	VESDA-E Filter	VSP-969-S	VESDA-E VES-A10-P Front Cover - Plastic - LCD - 3.5" Display
VSP-962-20	VESDA-E Filter - 20 Pieces	VSP-969-04-S *	VESDA-E VES-A10-P Front Cover - Plastic - LCD - 3.5" Display
VSP-963	VESDA-E Aspirator		

* Spare parts for GA4 only.

3.5" Display



Home Page



Sector Status Page

LED	Description
	Fire 2
	Fire 1
	Action
	Alert
	Disabled
	Fault
	Power

Home Page

Icon on Display	Description
	Smoke Level and Adaptive Scan Threshold
	Detector OK
	Detector Fault
	Aspirator Fault
	Airflow Fault
	Power Fault
	Filter Fault
	Smoke Chamber Fault
	VESDAnet Fault
	StaX Module Fault

Sector Status Page



Display Element	Description
	Sector Alarm Level
	Sector Smoke Level bargraph including alarm threshold indicators
	User-configured Sector Name

Approvals Compliance

Please refer to the Product Guide for details regarding compliant design, installation and commissioning.

www.xtralis.com

Doc. No. 33728_08
Part No. AD30909-003
July 2024

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TRIO-UPS-2G/1AC/24DC/10

Uninterruptible power supply



Data sheet
107432_en_03

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1 Description

The TRIO-UPS-2G uninterruptible power supply (UPS) with integrated power supply unit is used to ensure that secure DC power for critical electrical loads continues to be supplied in the event of disturbances in the power supply network, e.g. due to mains breakdown or failure. In the event of such a disturbance, the UPS switches to battery operation without interruption so that connected loads continue to be continuously supplied. In this operating mode, the output voltage is directly dependent on the battery voltage. When mains power is restored, the UPS automatically returns to normal operation. The connected loads are again supplied via the power supply network and the battery is charged at the same time. The TRIO-UPS-2G uninterruptible power supply (UPS) is designed specifically for supplying industrial PCs (IPCs).

Features

- Power supply, loading unit and electronic switchover unit in a device
- Wide-range input AC
- Adjustable output voltage in mains operation
- USB interface for configuration and diagnostics
- Adjustable charging parameters for the use of different types of energy storage
- Comprehensive signaling

Technical data (short form)

Input voltage range	100 V AC ... 240 V AC (-15 % ... +10 %) 110 V DC ... 250 V DC
Derating	< 90 V AC (2.5 %/V)
Frequency range (f_N)	50 Hz ... 60 Hz (± 10 %)
Typical current consumption (for nominal values)	4.5 A (100 V AC) / 2 A (240 V AC) 4 A (110 V DC) / 1.8 A (250 V DC)
Nominal output voltage (U_N)	24 V DC
Output current I_N / I_{Dyn}	10 A / 15 A
Residual ripple	< 20 mV
Efficiency (for nominal values) ($I_{CHARGE} = 0$; $I_{OUT} = I_N$)	typ. 90 % (120 V AC) / typ. 91 % (230 V AC)
MTBF (IEC 61709, SN 29500)	> 2007013 h (230 V AC, at 25 °C) > 1210518 h (230 V AC, at 40 °C) > 575978 h (230 V AC, at 60 °C)
Ambient temperature (operation)	-25 °C ... 70 °C (> 60 °C Derating: 2.5 %/K)
Ambient temperature (start-up type tested)	-40 °C
Dimensions (W/H/D)	68 mm / 130 mm / 160 mm
Weight	1.34 kg
Battery technology	VRLA-AGM
Nominal voltage (energy storage)	24 V DC
Charging current preset max.	2.1 A (-25 °C ... 40 °C) 3 A
Nominal capacity range	1.2 Ah ... 40 Ah
Capacitance preset	7 Ah



All technical specifications are nominal and refer to a room temperature of 25 °C and 70% relative humidity at 2000 m above sea level.

3 Ordering data

Description	Type	Item no.	Pcs./Pkt.
TRIO UPS - UPS with integrated power supply for lead AGM energy storage 1.2 Ah - 40 Ah, USB (Modbus/RTU), DIN rail mounting, Push-in connection, input: 1-phase, output: 24 V DC / 10 A	TRIO-UPS-2G/1AC/24DC/10	2907161	1
Accessories	Type	Item no.	Pcs./Pkt.
Energy storage, VRLA-AGM, 24 V DC, 1.2 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/1.2AH	1274520	1
Energy storage, VRLA-AGM, 24 V DC, 4 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/4AH	1274117	1
Energy storage, VRLA-AGM, 24 V DC, 7 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/7AH	1274118	1
Energy storage, VRLA-AGM, 24 V DC, 12 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/12AH	1274119	1
Energy storage, VRLA-AGM, 24 V DC, 20 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/20AH	1348516	1
Energy storage, VRLA-AGM, 24 V DC, 40 Ah, automatic detection and communication with QUINT UPS-IQ	UPS-BAT/PB/24DC/40AH	1354641	1
Energy storage device, lead AGM, VRLA technology, 24 V DC, 13 Ah, tool-free battery replacement, automatic detection, and communication with QUINT UPS-IQ	UPS-BAT/VRLA-WTR/24DC/13AH	2320416	1
Energy storage device, lead AGM, VRLA technology, 24 V DC, 26 Ah, tool-free battery replacement, automatic detection, and communication with QUINT UPS-IQ	UPS-BAT/VRLA-WTR/24DC/26AH	2320429	1
Used for communication between an industrial PC and Phoenix Contact devices with USB-Mini-B connection.	MINI-SCREW-USB-DATACABLE	2908217	1
2-piece universal wall adapter for securely mounting the device in the event of strong vibrations. The profiles that are screwed onto the side of the device are screwed directly onto the mounting surface. The universal wall adapter is attached on the left/right.	UWA 130	2901664	1
Universal wall adapter for securely mounting the device in the event of strong vibrations. The device is screwed directly onto the mounting surface. The universal wall adapter is attached on the top/bottom.	UWA 182/52	2938235	1



You will find the latest accessories for the item at [phoenixcontact.net/products](https://www.phoenixcontact.net/products).

4 Technical data

Input data

Input voltage range	100 V AC ... 240 V AC (-15 % ... +10 %) 110 V DC ... 250 V DC
Derating	< 90 V AC (2.5 %/V)
Frequency range (f_N)	50 Hz ... 60 Hz (± 10 %)
Typical current consumption (for nominal values)	4.5 A (100 V AC) / 2 A (240 V AC) 4 A (110 V DC) / 1.8 A (250 V DC)
Inrush current limitation (at 25°C)/ I^2t , typical	< 9 A / < 0.2 A ² s
Mains buffering (battery mode deactivated), typ.	≥ 25 ms (120 V AC) / ≥ 25 ms (230 V AC)
Input fuse slow-blow, internal	6.3 A
Permissible backup fuse	B10

Connection data: Input

Connection method	Push-in connection
Stripping length	10 mm
1-conductor rigid	0.2 mm ² ... 4 mm ²
1-conductor flexible	0.2 mm ² ... 2.5 mm ²
1-conductor flexible with ferrule without plastic sleeve	0.25 mm ² ... 2.5 mm ²
1-conductor flexible with ferrule with plastic sleeve	0.25 mm ² ... 1.5 mm ²
1-conductor rigid (AWG)	24 ... 12

Output data

Derating	> 60 °C (2.5%/K of P_{Out} nom.)
Crest factor	1.5 (120 V AC) / 1.5 (230 V AC)
Switch-over time	< 20 ms
Residual ripple	< 20 mV

Output data (mains operation)

Nominal output voltage (U_N)	24 V DC
Setting range (> 24 V constant capacity)	24 V DC ... 28 V DC
Output current I_N / I_{Dyn}	10 A / 15 A
Power dissipation	
No load	3 W
Nominal load	32 W

Output data (battery operation)

Nominal output voltage (U_N)	$U_{BAT} - 0.1$ V DC
Output voltage range	18 V DC ... 30 V DC
Output current I_N / I_{Dyn}	10 A / 15 A

Connection data: Output

Connection method	Push-in connection
Stripping length	10 mm
1-conductor rigid	0.2 mm ² ... 4 mm ²
1-conductor flexible	0.2 mm ² ... 2.5 mm ²
1-conductor flexible with ferrule without plastic sleeve	0.25 mm ² ... 2.5 mm ²
1-conductor flexible with ferrule with plastic sleeve	0.25 mm ² ... 1.5 mm ²
1-conductor rigid (AWG)	24 ... 12

Charging process

Battery technology	VRLA-AGM
Charge characteristic curve	IU ₀ U
Nominal voltage	24 V DC
Nominal capacity range	1.2 Ah ... 40 Ah
Capacitance preset	7 Ah
max. (recommended)	140 Ah
Charging voltage	max. 30 V DC
Charging current configurable	0.2 A ... 3 A (-25°C ... 40°C)
reduced	3 A ... 0 A (40 °C ... 65 °C)
preset	2.1 A (-25°C ... 40°C)
max.	3 A

Connection data: Battery

Connection method	Push-in connection
Stripping length	15 mm
1-conductor rigid	0.2 mm ² ... 10 mm ²
1-conductor flexible	0.2 mm ² ... 6 mm ²
1-conductor flexible with ferrule without plastic sleeve	0.25 mm ² ... 6 mm ²
1-conductor flexible with ferrule with plastic sleeve	0.25 mm ² ... 4 mm ²
1-conductor rigid (AWG)	24 ... 8

Status and diagnostic indicator / signal output DC OK

Connection labeling	3.1
Switching output	Transistor output, active
Output voltage	24 V DC
Continuous load current	20 mA
LED status indicator	green

Status and diagnostic indicator / signal output Alarm

Connection labeling	3.2
Switching output	Transistor output, active
Output voltage	24 V DC
Continuous load current	20 mA
LED status indicator	red

Status and diagnostic indicator / signal output Battery mode

Connection labeling	3.3
Switching output	Transistor output, active
Output voltage	24 V DC
Continuous load current	20 mA
LED status indicator	yellow

Status and diagnostic indicator / signal output Ready

Connection labeling	3.4
Switching output	Transistor output, active
Output voltage	24 V DC
Continuous load current	20 mA

Remote control Remote**Digital control input (configurable)**

Connection labeling	3.5
Low signal	Connection to SGnd with < 2.7 kΩ
High signal	Open (> 35 kΩ between Remote and SGnd)

Battery-operated start (Bat.-Start)**Low active digital control inputs**

Connection labeling	3.6
Low signal	Connection to SGnd with < 2.7 kΩ
High signal	Open (> 200 kΩ between Bat.-Start and SGnd)

Connection data: Signal

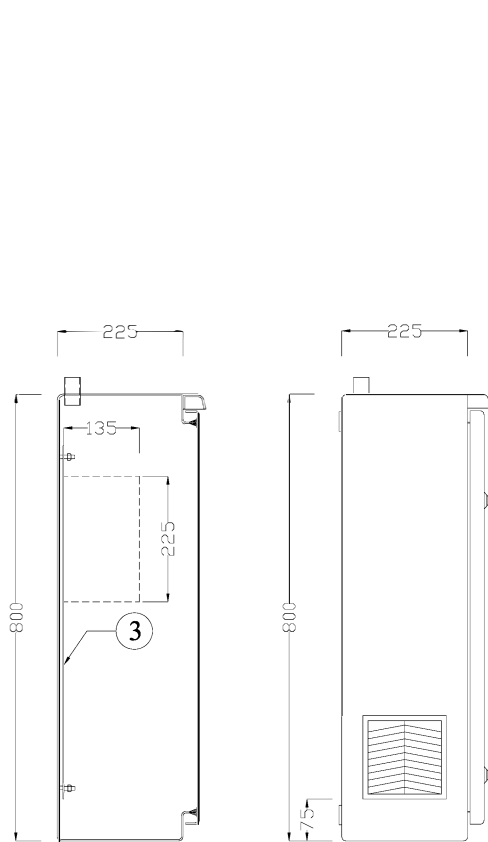
Connection method	Push-in connection
Stripping length	8 mm
1-conductor rigid	0.2 mm ² ... 1.5 mm ²
1-conductor flexible	0.2 mm ² ... 1.5 mm ²
1-conductor flexible with ferrule without plastic sleeve	0.2 mm ² ... 1.5 mm ²
1-conductor flexible with ferrule with plastic sleeve	0.2 mm ² ... 0.75 mm ²
1-conductor rigid (AWG)	24 ... 16

Data interface

Interface designation	USB (Modbus/RTU)
USB classification	MINI-USB Type B
Locking	Screw

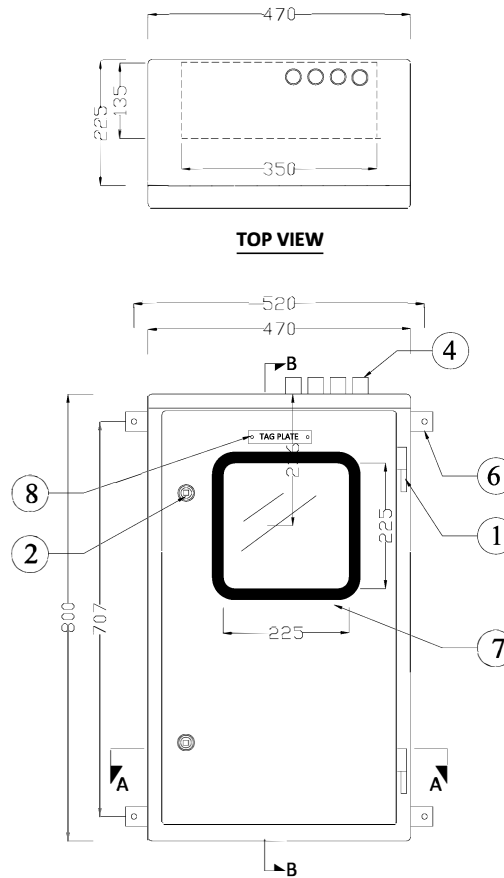
Device combinations

UPS connection in parallel	yes, with diode module uncoupled
UPS connection in series	no
Energy storage device connection in parallel	yes max. 3

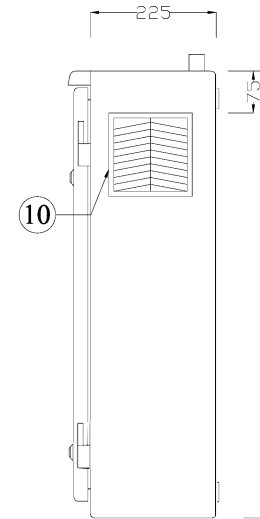


SECTION B-B

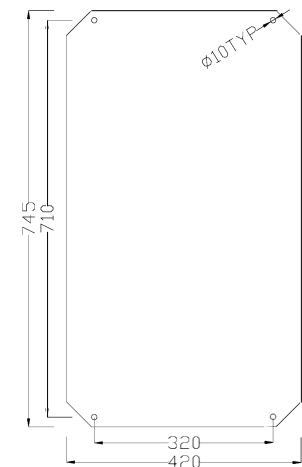
SIDE ELEVATION



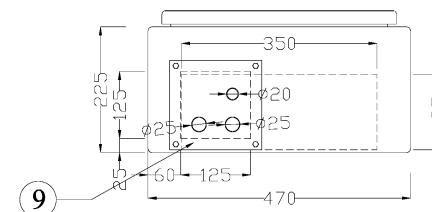
FRONT ELEVATION



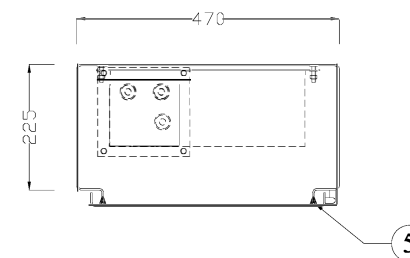
SIDE ELEVATION



3. Mounting Plate Details



BOTTOM VIEW



SECTION A-A

Include the approvals and COMPANY approval for the proposed cabinet.

DETAILS:

1. Liftoff hinges, SS316.
2. Quarter turn square key lock, SS316.
3. Mounting Plate, 1.5mm thk. GI, size: 745H x 420W mm (to mount Detection Panel).
4. HSSD socket molded on FRP wall to suits pipe entry (4 nos. on top, HSSD socket to be supplied by Tyco).
5. Bulb type EPDM rubber gasket for door sealing.
6. Wall mounting bracket to suit M10 bolts, 4 nos. (SS316).
7. UV Resistant glass window 225x225 mm, fixed by EPDM rubber gasket.
8. Tag/name Plate in SS316.
9. Gland Plate, 2mm thk. SS316. (holes to be made as per Tyco advise)
10. ABS louvers for natural ventilation, 2 nos. Civaux Part# CPF15.EX.G/W.

NOTES:

- a. All dimensions in millimeters.
- b. Enclosure material: FRP/GRP (Fiber Glass Reinforced Polyester).
- c. External finish-UV resistant gray color gelcoat finish to RAL 7035.
- d. Internal finish- flow coated finish to gray color.
- e. General wall thickness 4mm & 6mm at stress points.
- f. Ingress protection (as per BS EN 60529)
- g. Impact strength to IK10 (as per IEC 62262)
- h. Fire retardant & self extinguishing (as per BS 476: 7: class 2).
- h. Door aperture 700H x 370W mm.
- i. Dimension tolerance ± 10 mm.

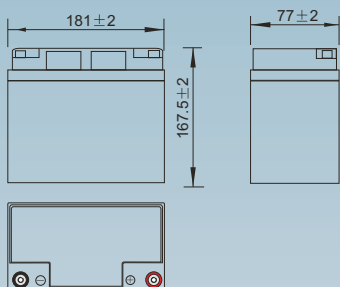
What is the purpose of the proposed cabinet, is it for containing the power supply and batteries or HSSD system and accessories, clarify.

01.	Gland Plate Hole & Viewing Window Location Revised	10/05/2023	M.M.
REV.	DESCRIPTION	DATE	CHECKED BY
First Reinforced Polyester Industries LLC P.O. Box No. 1621, U.A.Q, UAE, Tel. +971 6 539 0567 Email: sales@frp-industries.com, www.frp-industries.com			
PROJECT:-			
TITLE:- GENERAL ELEVATION			
SIZE:- 800H x 470W x 225D mm (Internal)			
DRG. NO:- FRP016-FES080047022B			
SCALE N.T.S.	DATE 26.01.2023	DRAWN BY O. V.	APPROVED BY D. N.

PG Range

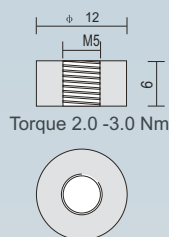
(6yr Design Life in Float Service at 20°C)

Dimensions (mm)



Standard terminal (mm)

T12 Threaded Insert 5mm Stud



Physical Specification

Nominal Voltage		12V
Nominal Capacity		21 AH
Dimension	Length	181 ± 2 mm
	Width	77 ± 2 mm
	Container Height	167.5 ± 2 mm
	Total Height (with terminal)	167.5 ± 2 mm
Weight		Approx 6.0kg
Standard Terminal		T12
Case and Cover Material		ABS (UL94-HB)

PG12V21



Operating Temperature Range

Storage (In fully charged condition)	- 20°C to + 60°C
Charge	- 15°C to + 50°C
Discharge	- 20°C to + 60°C

Discharge Characteristics (25°C, 77°F)

Rated Capacity	20hr Rate(1.80V/cell) 10hr Rate(1.80V/cell) 5hr Rate(1.75V/cell) 3hr Rate(1.75V/cell) 1hr Rate(1.60V/cell)	18.9AH 18.0AH 16.0AH 14.0AH 11.0AH	Cycle	Initial Charging Current Less than 5.0A Voltage 14.40V ~ 15.0V at 25°C (77°F)
Capacity Affected by Temperature	40°C (104°F) 25°C (77°F) 0°C (32°F)	103% 100% 86%		Recommended Charge Current less than 5.0A Voltage 13.5V ~ 13.8V at 25°C (77°F)

Internal Resistance mΩ

15.0 mΩ

Constant Current Discharge (Amperes) at 25°C (77°F)

F.V(V/cell)	15min	20min	30min	45min	1H	2H	3H	4H	5H	8H	10H	20H
1.85	20.6	17.1	13.3	10.5	8.52	5.58	4.19	3.43	2.90	2.03	1.72	0.916
1.80	22.8	18.8	14.3	11.2	8.98	5.94	4.42	3.60	3.05	2.13	1.80	0.945
1.75	25.3	20.6	15.4	12.0	9.69	6.22	4.68	3.76	3.20	2.19	1.84	0.965
1.70	27.7	22.5	16.9	12.5	10.20	6.56	4.89	3.91	3.33	2.27	1.90	0.984
1.65	29.3	23.8	17.9	13.3	10.60	6.79	5.07	4.05	3.40	2.33	1.94	1.007
1.60	32.1	25.8	19.0	13.7	11.00	7.07	5.24	4.18	3.52	2.39	1.98	1.031

Constant Power Discharge (Watts/cell) at 25°C (77°F)

F.V(V/cell)	15min	20min	30min	45min	1H	2H	3H	4H	5H	8H	10H	20H
1.85	38.8	32.6	25.6	20.4	16.7	11.0	8.24	6.77	5.74	4.04	3.45	1.83
1.80	42.5	35.3	27.2	21.4	17.4	11.6	8.65	7.06	6.00	4.23	3.59	1.89
1.75	46.5	38.3	29.0	22.8	18.7	12.1	9.12	7.35	6.29	4.34	3.67	1.93
1.70	50.1	41.5	31.7	23.7	19.7	12.7	9.52	7.65	6.53	4.50	3.78	1.96
1.67	52.9	43.7	33.3	25.1	20.3	13.1	9.85	7.90	6.66	4.61	3.86	2.01
1.60	56.8	46.7	35.0	25.7	20.9	13.5	10.11	8.11	6.86	4.72	3.94	2.05



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sales@power-sonic.co.uk
www.power-sonic.co.uk

VESDA[®]
by  **xtralis[®]**

Aspirating Pipe
and Fittings

 **xtralis[®]**
The sooner you know[®]

With over 250,000 aspirating smoke detectors installed world wide the VESDA name has become synonymous with high quality and outstanding performance.

An aspirating smoke detection system, however, only performs at its best with a well designed and installed quality pipework system. Xtralis, manufacturer of VESDA detectors, recognised this requirement and in 1997 launched its VESDA Pipe and Fittings range. Since then we have been providing pipes and fittings for customers in over 25 countries and the list of satisfied clients becomes longer every day. Now we can share the secret of our success.

Quality does not necessarily cost more

We provide a one-stop source for aspirating detector, pipe, fittings and accessories thus saving you procurement time and cost.

We chose to manufacture our products from ABS (Acrylonitrile Butadiene Styrene), a very lightweight, versatile material, guaranteeing you easy and professional installation.

In addition to our high quality products we are committed to providing unparalleled customer services including Fast Track ordering, next day delivery in the U.K. mainland and outstanding technical support.

Designing pipework has never been easier

We have developed a computer software programme for you to design and model pipe system layouts so that you can apply the optimum solution for each system application. The VESDA ASPIRE2 software helps you to estimate:

- Smoke transport time
- Balance between air-flows amongst sampling pipes and holes
- Influence of varying end cap vent sizes
- Effect of various fittings
- Classification to EN54-20 (Class A, B or C).

You can attend one of our regular VESDA Accreditation Training Courses to become a confident user of the VESDA ASPIRE2 software.

We have also produced Application Design Guides to help you avoid design and installation problems and optimise the performance of your aspirating smoke detection system.

In addition to our free software tool and design guides we provide you with committed customer service – our experienced field engineers give you full technical advice and support on design and installation. Should you require our services for commissioning assistance or system performance testing our technical staff are always ready to help you.

VESDA by Xtralis...the sooner you know

VESDA Aspirating Pipe and Fittings are produced in red ABS under a stringent quality control system approved to BS EN ISO 9001, EN54 - 20 and EN61386-1 Class 1131 ABS (Acrylonitrile Butadiene Styrene) is a robust polymer, which has the most suitable physical properties for aspirating detection, optimising performance with tensile strength, chemical resistance, ductility, weatherability, heat stability and processability.

Toughness and durability

The Butadiene constituent of ABS enhances impact strength and toughness at temperatures ranging from -40°C to 70°C and gives exceptional resistance to accidental damage*.

Chemical resistance

Acrylonitrile impacts chemical resistance, thus ABS is unaffected by corrosive attack from a wide range of acids, alkaline and other aggressive chemicals.

Easy to joint

The Styrene constituent in ABS makes it an easy material to joint using ABS solvent. Simple modifications to existing pipework are also possible with minimal specialist training.

Lightweight

Unlike some metal pipes, VESDA ABS pipework is very light in weight. This makes it easy to handle in high and difficult places where systems are often installed.

Color

The standard VESDA pipe and fittings are supplied in fire red to identify its service as fire. Special orders of white can be accommodated for aesthetic reasons.

*Pipe work must be installed to accommodate thermal expansion.

Our range of VESDA pipe and fittings

Our range of Pipe and Fittings products have been specially developed to provide fully matched components for the optimal performance of your whole VESDA aspirating system. They are also manufactured to comply with the requirements of BS5391 Part 1, EN 61386-1 and the new EN54 Part 20 guaranteeing you the quality you are used to as a VESDA customer.

The pipes are a metric standard of 25 mm external diameter with suitable adapters available for imperial to metric conversions and are designed for use with any make of low pressure aspirating type smoke detection system.

Some pipes and fittings are not available in certain countries, please check with an Xtralis office before you order.

GREEN Highlighting by Vendor, the highlights by Vendor are not considered to be comments, highlighting merely indicates the proposed type/model of the parts.

	Pipe PIP-001
	Socket PIP-002
	Socket Union PIP-003
	Adaptor 25 mm-³/₄" PIP-004
	90 Long Radius Bend PIP-005
	45 Elbow PIP-006
	End Cap PIP-007
	Equal Tee PIP-008
	Pipe Clip PIP-009
	Solvent Cement PIP-012
	Pipe Cutters PIP-014
	Flush Sample Point - Head only PIP-015
	Trunk Adapter PIP-016

	Flush Sample Point Kit 059-001
	Conical Sample Point Kit 059-007
	Label-Sampling Point 128-014
	Label-Smoke Detector Pipe 128-015
	Label-Sampling Point Wrap Around 128-046
	Conical Sample Point - Head only 144-013
	10 mm OD Capillary Pipe 221-035
	Discreet End Cap (clear) 222-059
	Smoke Test Wire 251-001
	Smoke Matches 251-002
	Smoke Pellets 251-003
	Mounting Base for sprinkler sampling point E700-SRB-SH
	Hot Wire Test Box VTT-10000